C-Reactive Protein (CRP), Quantitative

Test Details

METHODOLOGY

Latex immunoturbidimetry

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Quantitative concentration of CRP (mg/L) in serum

Use

CRP is an acute phase reactant, which can be used as a test for inflammatory diseases, infections, and neoplastic diseases. Progressive increases correlate with increases of inflammation/injury. CRP is a more sensitive, rapidly responding indicator than ESR. CRP may be used to detect early postoperative wound infection and to follow therapeutic response to anti-inflammatory agents. Recent reports have indicated that a highly sensitive version of the CRP assay may be used as an additional indicator for susceptibility to cardiac disease.

Limitations

CRP arises as a nonspecific response to tissue injury and inflammation.

Custom Additional Information

CRP is a pentameric globulin with mobility near the γ zone. It is an acute phase reactant which rises rapidly, but nonspecifically in response to tissue injury and inflammation. It is particularly useful in detecting occult infections, acute appendicitis, particularly in leukemia and in postoperative patients. In uncomplicated postoperative recovery, CRP peaks on the third postop day, and returns to preop levels by day seven. It may also be helpful in evaluating extension or reinfarction after myocardial infarction, and in following response to therapy in rheumatic disorders. It may help to differentiate Crohn's disease (high CRP) from ulcerative colitis (low CRP), and rheumatoid arthritis (high CRP) from uncomplicated lupus (low CRP).

Specimen Requirements

Specimen

Serum

Volume

1 mL

Container

Red-top tube or gel-barrier tube

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Male (mg/L)	Female (mg/L)
0 to 30 d	Not established	Not established
1 m to 17 y	0–7	0–9
>17 y	0–10	0–10

Storage Instructions

Room temperature

Causes for Rejection

Gross hemolysis; lipemia